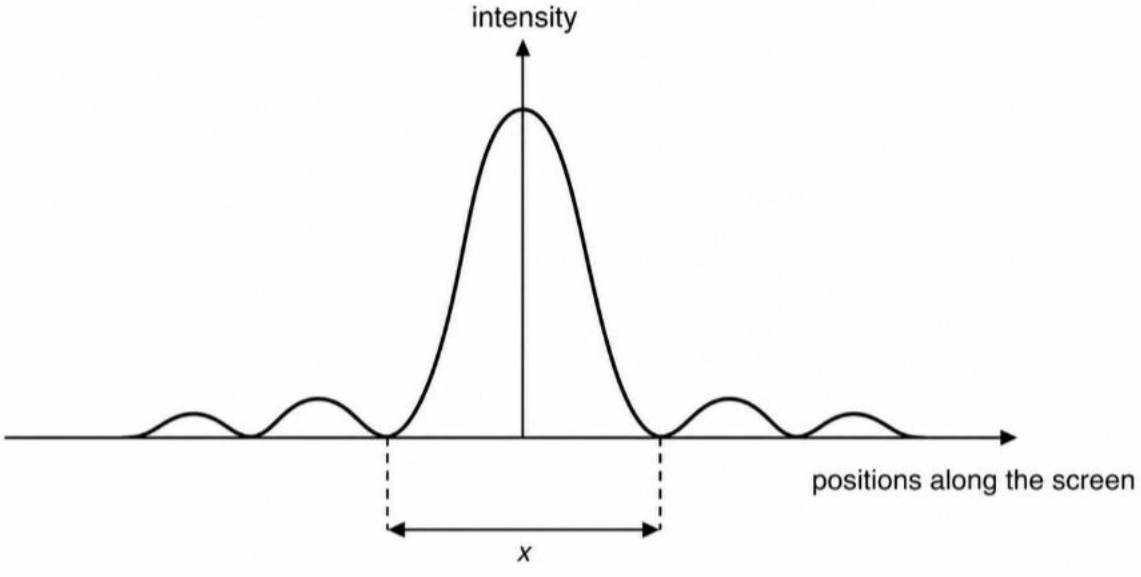
14 A parallel beam of monochromatic light passes through a narrow slit and forms a diffraction pattern on a distant screen. The width x is the distance between the first minima on either side of the central maximum.



Which statement is **incorrect**?

A Moving the screen closer to the slit increases x.

B Increasing the slit width generally allows more light to pass through the slit.

C Increasing the slit width decreases x.

D Increasing the wavelength increases x.

Ans: A

Single slit equation: $b\sin\theta = \lambda$
For small angles:

$$\theta \approx \frac{\lambda}{b}$$

Using the geometry of the screen and slit, the width of the central maximum on a distant screen is approximately

$$x = 2D\theta \approx \frac{2D\lambda}{b}$$

Option A: Making D smaller means x will be smaller.
Option B: That's obviously true as a bigger slit allows more light to pass through.
Option C: Increasing b makes x smaller.
Option D: Increasing λ makes x smaller.